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Long short-term memory neural network for directly inverse design of nanofin metasurface

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In this Letter, the neural network long short-term memory (LSTM) is used to quickly and accurately predict the polarization sensitivity of a nanofin metasurface. In the forward prediction, we construct a deep neural network (DNN) with the same structure for comparison with LSTM. The test results show that LSTM has a higher accuracy and better robustness than DNN in similar cases. In the inverse design, we directly build an LSTM to reverse the design similar to the forward prediction network. By inputting the extinction ratio value in 8–12 μm, the inverse network can directly provide the unit cell geometry of the nanofin metasurface. Compared with other methods used to inverse design photonic structures using deep learning, our method is more direct because no other networks are introduced. © 2022 Optica Publishing Group

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The fast and accurate acquisition of the electromagnetic response of photonic structures can aid in the development and design of optical devices. Computer simulations can help us understand the characteristics of optical devices without experiments, significantly reducing the research time and cost. Most electromagnetic simulations are based on full-wave simulation algorithms such as the finite difference time domain (FDTD) and finite element method (FEM). The simulation of these algorithms requires considerable computing resources and is time-consuming. In recent years, the success of artificial intelligence algorithms based on deep learning in image processing, speech recognition, natural language processing, and other fields has led many researchers to realize that deep learning can be used in the design of photonic structures, providing a new idea for their design [1,2]. By inputting a large amount of data to train the neural network, the trained network can quickly and accurately provide an optical response. Compared to traditional algorithms, this method is considerably faster in obtaining an optical response. Usually, using a neural network is 10^4 times faster than using a full-wave simulation algorithm. Additionally, deep learning has made the task of designing photonic structures more intelligent. The size, structure, and other parameters of the device can be determined directly by the input electromagnetic response. The former task is often referred to as forward prediction and the latter is the inverse design. In the forward prediction, the accuracy of prediction is of great concern to researchers. In the inverse design, there is a non-uniqueness problem, because the electromagnetic response often corresponds to multiple combinations of photonic sizes or structures. To solve the non-uniqueness problem of the deep learning inverse design of photonic structures, many models have been proposed, including tandem networks [3–7], cGAN [8], and auto-encoders [9]. However, many of these methods did not directly realize the inverse design but instead inserted another network or vector into the constraint output parameters. For example, a pre-trained forward prediction network is required in tandem networks, both generation and adversation networks are required in cGAN, and encoding and decoding networks are required in auto-encoders.

A metasurface, a human-made material, has many novel characteristics that do not exist in nature [10–12]. By designing a subwavelength structure, optical characteristics such as the amplitude, phase, and polarization can be manipulated. Many metasurface-based devices have been manufactured, including the metalens [13], wave plate [14], beam splitter [15], holography [16], and many others [17]. The design of metasurfaces based on deep learning has also attracted extensive attention [18]. Moreover, the design of metasurfaces using deep learning is intelligent and efficient. It does not require much personal experience or trial and error [19]. Most of these deep-learning-based metasurface design works focus on the spectrum, phase, and far-field responses [20,21]. Polarization is also a very important characteristic of electromagnetic waves. Owing to the different dimensions of the orthogonal axis, many subwavelength metasurface devices are polarization sensitive. While two orthogonally polarized incidences propagate through the device, there is polarization sensitivity. Most intuitively, the transmittances are different, whereas the polarization sensitivity exists at a certain wavelength. The extinction ratio (ER) [22] can be used to quantify the degree of polarization sensitivity.

Traditional neural networks assume that inputs and outputs are independent. Sometimes the data are interrelated. Recurrent neural networks (RNNs) are useful for dealing with sequential data. The basic principle of an RNN is to save output and feedback to the input. When the sequential data increases, the RNN cannot connect with the data far from it. To overcome this issue, a new module referred to as long short-term memory (LSTM)

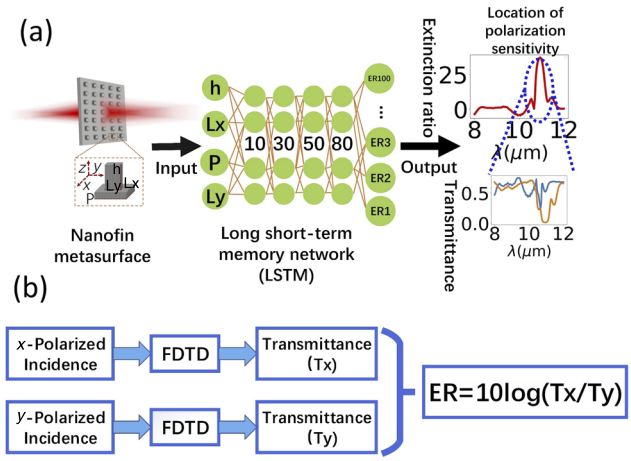

Fig. 1. (a) LSTM as predicting network; (b) process of calculating the ER.

networks was proposed. LSTM is a special type of RNN that is also a chain module capable of learning long-term sequential data (see Supplement 1). In this study, the ER was used to train the LSTM neural network to design a nanofin metasurface [23]. Nanofin metasurfaces have simple structures and the unit cell consists of several degrees of freedom and can be easily manufactured. Many optical device-based nanofin metasurfaces have been proposed [24]. There is some hidden relevance between four input geometric parameters of a nanofin-based metasurface and its polarization response. For example, the change of length along the x axis not only causes x-polarization response shift, but also leads to the change of y-polarization response. The LSTM may learn this connection information. The trained LSTM can provide ER values quickly and accurately. We compared it with a deep neural network (DNN), the most commonly used network in deep learning design photonic structures, and found that LSTM has higher accuracy and better robustness under the same conditions. We also designed an LSTM to realize the design of the nanofin metasurface. During the inverse design, the ER values were directly taken as the input, and the geometric parameters of the unit cell for the nanofin metasurface were taken as the output to train the inverse network. Unlike other approaches, our inverse design method does not include another network or vector. It directly input the geometric parameters and output ER values. Our inverse design network is therefore simpler and more straightforward.

The forward prediction method is illustrated in Fig. 1(a). The nanofin metasurface is made of silicon, and its refractive index is 3.42 in the 8–12 μm spectrum. P, Lx, Ly, and h represent the pitch, length along the x axis, length along the y axis, and height of the nanofin metasurface unit cell, respectively. These four geometric parameters were input to the network and the network output ER values. Figure 1(b) shows the calculation of the ER values. The ER is defined as

$$\mathrm{ER} = 10 * \log(\mathrm{Tx}/\mathrm{Ty}). \tag{1}$$

Tx and Ty are the corresponding transmittances for the x- and y-polarized incidences, respectively.

It can be deduced from Eq. (1) that the ER values can reflect the degree of polarization sensitivity. Ideally, if the ER is equal to 0, there is no polarization sensitivity. After calculating the ER by simulation and calculation, 100 points were uniformly sampled in the 8–12 μm spectrum as the output.

The LSTM used in this study contained six layers, including one input layer, one output layer, and four hidden layers. As shown in Fig. 1(a), the four hidden layers contained 10, 30, 50, and 80 cells, respectively. Moreover, the last layer is a linear connection that ensures that the network has a fast convergence speed during training (see Supplement 1). We generated 8000 different combinations of groups of geometric parameters for an all-silicon unit cell and used the FDTD method to obtain the transmittances. Then, we used Eq. (1) to calculate the ER. These 8000 groups of data were used for training. The geometric normalizing factor was set to 10 μm so that most of the input data values were between 0 and 1.

Loss function is used during training which is defined as

$$\mathrm{L_{ER}} = \frac{\sum_{i}^{n} |\mathrm{ER_{sim}} - \mathrm{ER_{pre}}|}{n}, \tag{2}$$

where $\mathrm{ER_{sim}}$ is the extinction ratio calculated using simulated data and $\mathrm{ER_{pre}}$ is the extinction ratio predicted by LSTM. The AdamOptimizer and learning rate 10^{-2} are used. For LSTM, an input sequence of our four parameters may affect the predicted results. To choose the best sequence, we used the same training datasets to evaluate the characteristic of 24 input sequences (see Supplement 1). Here, 200 groups of test data are generated, with significant differences among them and no overlap with training data. The mean value μ and standard deviation σ are used to quantitatively evaluate the test results. The smaller the mean μ is, the better the prediction results. Moreover, the smaller the standard deviation σ, the better the robustness of network prediction. The μ and σ are defined as

$$\mu = \frac{1}{200}\sum_{i=1}^{i=200}\{(\frac{1}{100}\sum_{j=1}^{j=100}|\mathrm{ER}_{\mathrm{true}}^{j} - \mathrm{ER}_{\mathrm{pre}}^{j}|\}), \tag{3}$$

$$\sigma = \sqrt{\left\{\frac{1}{200}\sum_{i=1}^{i=200}\left[\left(\frac{1}{100}\sum_{j=1}^{j=100}|\mathrm{ER}_{\mathrm{true}}^{j} - \mathrm{ER}_{\mathrm{pre}}^{j}|\right) - \mu\right]^{2}\right\}}, \tag{4}$$

where $\mathrm{ER_{true}}^{j}$ is the true jth ER value in the 8–12 μm obtained by the FDTD method and $\mathrm{ER_{pre}}^{j}$ is the jth ER value predicted by LSTM. Figure 2(a) shows the μ-σ distribution for all test results. In the μ-σ distribution in Fig. 2(a), the blue dot represents the DNN and the red dots represent the LSTM for different input sequences. Detail information is shown in Table S1 in Supplement 1. It can be observed that the sequence 'h, Lx, P, Ly', plot with the green dot in Fig. 2(a), has the minimum values for μ and σ, which means that this sequence has the best prediction results in the comparison. A DNN is the most commonly used network for photonic structure design in deep learning [1]. We also constructed a DNN for comparison with the LSTM. The number of layers and unit cells in each layer was the same as that of the LSTM. Other conditions, such as

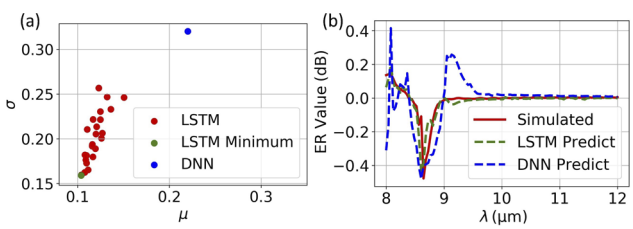

Fig. 2. (a) μ–σ distribution of LSTM versus DNN; (b) example.

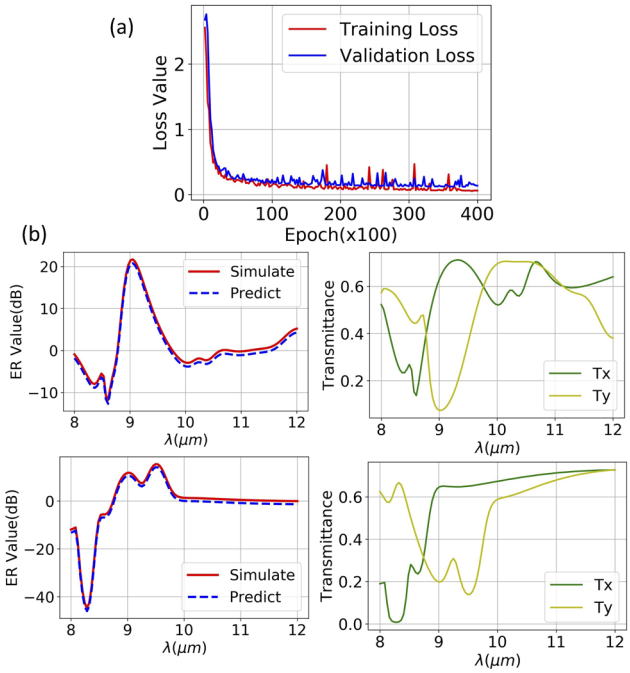

Fig. 3. (a) Loss values: training and validation; (b) two examples for predicting and simulated results.

the learning rate and optimization method, datasets including the training and test datasets, and training epochs, are also the same. It can be observed that the μ and σ values of the DNN are considerably larger than those of the LSTM. This means that under the same conditions, the accuracy and robustness of the DNN are worse than those of the LSTM. Figure 2(b) shows an example in which the DNN cannot predict accurately, whereas the LSTM can work successfully. The LSTM and different DNNs were also constructed and well-optimized. We compared them with each other. It shows that LSTM has higher accuracy in the predictions with 4 layers and 170 nodes. DNNs have the same prediction results while it has much greater number of layers and nodes than LSTM (see Supplement 1 in the Supplementary material). The test results show that LSTM is better than DNN in our study under the same conditions.

Based on the input sequence evaluation mentioned above, we chose the sequence 'h, Lx, P, Ly' as the input, as shown by the green dots in Fig. 2(a). Additionally, 8000 groups of data were used for training and 3000 groups were used for verification. Figure 3(a) shows the training and validation loss values. The values dropped rapidly after 1000 epochs of training, and after 40,000 epochs the training dropped to 0.06, whereas the verification loss values dropped to 0.08. A comparison between the two examples of the prediction and simulation results is shown in Fig. 3(b). The left column shows the ER values of the predicted and simulated results. The right column shows the relative transmittances for the *x*- and *y*-polarization incidences. The predicted results are in good agreement with the simulated results.

We also used ER values as the input and geometric parameters as the output to train the network to realize an inverse design, as shown in Fig. 4(a). The inverse network is similar to the prediction network; however, the inputs and outputs are transformed. It also contained six layers, including the input and output layers, and four hidden layers. The four hidden layers contained 80, 50, 30, and 10 cells, respectively. The last layer also had a

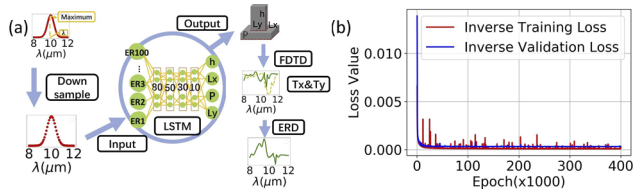

Fig. 4. (a) Schematic of inverse design (ERD: extinction ratio distribution); (b) loss values.

linear connection. The AdamOptimizer and a learning rate of 10^{-2} were proposed. The loss function is defined as follows:

$$L_{size} = \frac{\sum_i^n |S_{True} - S_{pre}|}{n}, \tag{5}$$

where S_{True} is a real geometric parameter vector and S_{pre} is the prediction of the geometric parameter vector by LSTM. Using the same 8000 groups of data as in the forward network, the accuracy is poor, unless the data increased to 30,000 for the inverse network. So, we generated 30,000 groups of data for training and 3000 groups of data for testing.

To test the ability of inverse-design network, we establish the following function:

$$ER(\lambda) = 25 * e^{-(x-\lambda)^2} + 0.1, \tag{6}$$

as input for the trained model, where x and λ are the selected and central wavelengths, respectively. The larger the value of ER, the stronger the polarization sensitivity and the larger the transmittance difference for two orthogonally polarized incidences. By inputting the ER values of the entire band from 8 to 12 μm into the trained network, the geometric parameters can be quickly obtained. Using these predicted geometric parameters, the designed nanofin metasurface had the strongest polarization sensitivity at a given maximum ER value.

Figure 5(a) shows an example of the input and predicted results. The red line in the left column is the input ER values, whereas the blue dotted line is the simulated ER values calculated by simulated transmittances using the predicted geometric parameters of the nanofin metasurface, which is simulated by the FDTD method. The right column shows the simulated transmittances by the FDTD method. The solid green line represents the transmittance for the *x*-polarized incidence and the dotted yellow

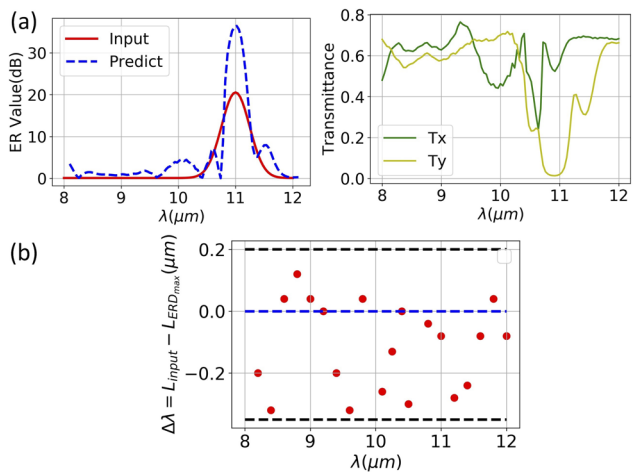

Fig. 5. (a) Example for input compared with predicted results; (b) test errors for 20 positions expanding 8–12 μm.

line represents the transmittance for the y-polarized incidence. In the first example, the input ER values follow the equation

$$\mathrm{ER} = 25 * e^{-(x-11.0)^2} + 0.1. \tag{7}$$

The predicted ER values reached their maximum around λ=11.0 µm. The inverse-design network yielded highly accurate prediction results.

To test the ability of the inverse design in the entire band of 8–12 µm, we tested 20 groups of data. The maximum positions of the ER values are distributed across the entire band of 8–12 µm. Figure 5(b) shows the test results for the difference between the input ER maximum positions and predicted ER maximum positions. It can be observed from Fig. 5(b) that the position differences of the maximum value are less than 0.3 µm, indicating that our inverse-design method has high accuracy in the entire band of 8–12 µm.

As a surrogate model, deep learning can quickly predict electromagnetic responses. This can be more than 10^4 times faster than that of the full-wave algorithm. Deep learning can be more convenient for predicting light responses, such as the transmittance, phase, and far-field of the photonic structure. Polarization sensitivity is an important characteristic often found in nanofin metasurfaces. The effective refractive index model is used to explain this problem while considering nanofin as a vertical waveguide. However, it cannot operate at all wavelengths and cannot be used for inverse design [25]. Previous studies on deep learning inverse-designing photonic structures often used multiple networks to constrain parameters because a completely inverse design is a non-uniqueness problem [26]. We used the ER as the input and realized a direct inverse design. This is more straightforward. Because the ER is an internal characteristic of nanofin metasurfaces, it presents a new method of inverse design for photonic structures. Other shapes of metasurface with polarization sensitivity are also suitable for this method in inverse design. Our methods are not only suitable for polarization-sensitive metasurfaces, but can also be used in other continuous optical response predictions, such as phase and spectral parameters with changes of nanostructure in the future.

In conclusion, we used an LSTM neural network to predict the polarization sensitivity of a nanofin metasurface. LSTM was much more accurate than DNN under the same conditions in this study. The ER, which reflects the polarization sensitivity of the metasurface, was used to train the network. Additionally, an inverse network was constructed. The ER was set as the input to train the LSTM. The trained network could quickly and accurately provide the geometric parameters of the nanofin metasurface. Our method showed that neural networks cannot only predict polarization sensitivity but can also realize inverse design using only one network. This provides a new method for deep learning in inverse-design photonic structures. Our work reveals that LSTM can be used to design nanophotonic structures with inverse-design capabilities.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

Supplemental document. See Supplement 1 for supporting content.

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